

When Consciousness Cannot Be Identified: A Descent-Theoretic Framework for Scientific Admissibility

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Abstract

Scientific attributions of consciousness require an identification claim: the target property must be recoverable from a declared observation site. This paper studies hidden-variable systems in which that recovery fails in a way that is stable under admissible revisions of the measurement language. The class is specified by local contextual presentability, nontrivial transition structure, nontrivial holonomy, and equivariant admissible operations. For such systems, the target property is non-identified relative to the declared site and revision envelope. Formally, the paper gives a revision-stable persistent non-effective descent criterion, conditional on a descent-factorization assumption proved here for the finite cases and isolated as an explicit obligation in general. Kernel-pair and conservative-invariance results are verified in Lean for the finite and categorical cases. Philosophically, it defends a bounded admissibility rule for consciousness attribution: a low-concern verdict requires identification evidence, not merely behavioral, access/report, or architectural evidence. Where the target is non-identified, or where the relevant structure is undisclosed, a low-concern verdict is not scientifically licensed. The paper applies the criterion to Integrated Information Theory, Global Workspace Theory, and Higher-Order Theory, and gives a worked dossier for Llama 3.1 8B-Instruct. The Llama dossier should be read narrowly: it is not a claim about phenomenal status. Its role is to show that access/report claims can be evaluated under a declared site while phenomenal-consciousness verdicts remain non-applicable without target, classifier, challenge, and provenance evidence.

Keywords: consciousness; scientific admissibility; identification; descent theory; artificial intelligence systems; philosophy of science

Technical summary

A hidden-variable system H generates revision-stable necessary non-identification of target property Θ , under the descent-factorization assumption, when the target fiber $F \rightarrow H$ has non-effective descent over the admissible observation site $\mathbb{C}_Y$ for every admissible structural revision $r \in \text{Rev}_\Omega(Y)$. The revision envelope $\text{Rev}_\Omega(Y)$ covers future structural changes to the observation framework, not only refinements within a fixed language. Theorem 1 strictly subsumes the kernel-pair obstruction lemma (Lemma 1) as the classical special case. Integrated Information Theory [Tononi et al., 2016], Global Workspace Theory [Baars, 1988, Dehaene and Changeux, 2011], and Higher-Order Theories [Rosenthal, 2005] illustrate three different outcomes.

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1 Identification Failure as a Structural Constraint

Imagine an AI company claims its model deserves a “low concern” verdict on moral status: there is no significant probability the system is conscious. What evidence would settle this? The company offers behavioral evidence (the system does not exhibit distress signals), functional evidence (it lacks the neural correlates of consciousness), and architectural evidence (its computation is dissimilar to human cognition). A regulator reads the evidence and asks whether it separates consciousness from an observational duplicate.

The difficulty lies in the relation between the target and the evidence. Behavioral competence can be duplicated by a system that lacks consciousness. Neural-correlate evidence is tied to human neural architecture. Architectural dissimilarity establishes difference, not absence. The proposed tests therefore leave two stories open, with no separating experiment yet on the table.

The philosophical literature calls this the “hard problem.” The present paper uses a more specific framing: it is an *identification problem*. A parameter is identified if different true values lead to different observable distributions. When identification fails, more data does not help.

The central question. Is identification failure for consciousness *necessary*, in the sense that it survives admissible improvements in measurement, or *contingent*, in the sense that a better admissible test could remove it? The distinction determines whether a failed test should prompt further measurement or a restricted verdict.

For a well-defined class of systems the failure is necessary relative to the declared observation site and revision envelope, conditional on the descent-factorization assumption stated in Theorem 1 and proved below for the finite deterministic and exact finite stochastic cases. The next sections characterize that class, state the descent criterion, and derive the governance consequence: these systems cannot receive an auditable low-concern verdict without a successful identification certificate.

2 Formal Preliminaries

The setup needs three ingredients. First, “admissible observations” names the measurements permitted for the system under evaluation. Second, experimental indistinguishability says that two hidden states agree under all of those observations. Third, the *experimentability witness* records the minimal typed specification a scientific consciousness claim must provide. A theory that cannot supply this witness is not falsifiable, and the theorems below do not apply to it.

Setup. Let H be the set of possible hidden states of a system (its internal state space: neurons firing, weights activating, whatever physical process underlies behavior). Let $\Theta : H \rightarrow T$ be the target map (e.g., “is this state conscious or not?”). Let Y be the collection of admissible observations: every measurement, perturbation, imaging technique, and behavioral test permitted. Let $E_Y : H \rightarrow \mathcal{R}_Y$ be the observation function mapping hidden states to observable records.

Definition 1 (When two hidden states look identical). *Hidden states h_1 and h_2 are experimentally indistinguishable if every admissible measurement gives the same distribution:*

$$h_1 \equiv_Y h_2 \iff \sup_{e \in \text{Eff}(Y)} |p(e \mid h_1) - p(e \mid h_2)| = 0.$$

On classical deterministic systems this simplifies to $E_Y(h_1) = E_Y(h_2)$.

Definition 2 (Experimentability witness). *A theory (H, Y, E_Y, Θ) is experimentable if it supplies a typed specification $W = (\mathcal{C}, \mathcal{R}, \mathcal{T}, H, Y, E_Y, \Theta)$ in which $\mathcal{R}$ and $\mathcal{T}$ have decidable equality. By*

“decidable equality” we mean a procedure that definitively answers “did these two records match?” and “did these two target values match?” If no such specification can be given, the theory is not making a falsifiable scientific claim in the sense used here.

Remark 1. A governance regulator presented with a theory that cannot supply the witness should require the witness before deliberating about the theory. Section 9 shows that IIT, GWT, and HOT occupy different positions under this test. IIT does not yet supply the witness for full Φ -based identification. GWT and HOT can supply witnesses for narrower targets.

3 Structural Characterization of the Non-Identification Class

The formal machinery in §§4–5 establishes when a hidden-variable system necessarily fails identification. Before that machinery, this section characterizes the class of systems it applies to. Four structural properties, stated without reference to consciousness, jointly specify which hidden-variable systems fall in the necessary non-identification class. A system satisfying all four properties will generate $Z = 1$ by the theorem in §5.

Definition 3 (P1, Local Contextual Presentability). A family of admissible local contexts $\{U_\alpha \rightarrow H\}$ exists such that each U_α supports local target assignments. The target Θ is partially evaluable at the level of individual contexts, even when not globally evaluable.

Definition 4 (P2, Nontrivial Transition Structure). On overlapping contexts $U_\alpha \cap U_\beta$, local target assignments are related by transition maps $g_{\alpha\beta} : F_\beta|_{U_{\alpha\beta}} \rightarrow F_\alpha|_{U_{\alpha\beta}}$ that satisfy local compatibility on every pair but fail global compatibility across all contexts simultaneously.

Definition 5 (P3, Nontrivial Holonomy). The context groupoid $\Pi_Y(H)$ acts on the target fiber F with no admissible global fixed section: $\text{Fix}(\rho) = \emptyset$. The descent datum is non-effective: no global section is compatible with all local data simultaneously.

Definition 6 (P4, Equivariant Admissible Operations). Every admissible intervention or observation refinement $Y' \in \text{Adm}(Y)$ preserves the holonomy structure: $\rho_{Y'} \circ a = a \circ \rho_Y$ for every admissible action a . No expansion or refinement of the observation family dissolves the obstruction.

P1–P3 characterize a static obstruction at one observation level. P4 is the persistence condition. It guarantees the holonomy survives every admissible intervention. A system missing P4 has a contingent obstruction, removable in principle by a better instrument. A system satisfying all four has a necessary obstruction, one that no improvement in measurement methodology can dissolve.

These properties depend only on the causal graph topology and the target fiber’s descent behavior over the observation site. They make no reference to substrate, material composition, or biological origin. Any physical system, biological or digital, that instantiates equivalent P1–P4 structure falls in the same identification class.

The claim is not that P1–P4 exhaust every possible source of the hard problem. The narrower claim is stability under conservative re-presentation. For systems with P1–P4, conservative expansions of the structural language produce no new structural distinctions and therefore cannot supply the sufficiency bridge from non-identification to consciousness attribution. Non-conservative expansions may matter, but they change the admissible site Y and must be declared in the revision envelope Ω . The boundary result below concerns that conservative-stability claim, not a complete taxonomy of all possible consciousness obstacles.

4 The Kernel-Pair Non-Factorization Lemma

The simplest way identification can fail is also the easiest to miss. Two hidden states look identical from outside (every measurement returns the same result) but differ in whether Θ holds. For consciousness: a conscious system and a non-conscious system produce identical observable records under every admissible test. No experiment distinguishes them. This is the kernel-pair obstruction.

Definition 7 (Kernel pair). *The kernel pair of E_Y is the set of all experimentally indistinguishable pairs:*

$$K_Y = \{(h_1, h_2) \in H^2 : h_1 \equiv_Y h_2\}.$$

Think of K_Y as the observation window's blind spot. The quotient $H/\sim_Y$ is the compressed state space in which indistinguishable states are treated as one.

Lemma 1 (Kernel-pair obstruction). *For an experimentable theory (H, Y, E_Y, Θ) :*

Θ is necessarily non-identified if and only if there exist hidden states h_1, h_2 that look identical from outside ($h_1 \equiv_Y h_2$) but differ in the target ($\Theta(h_1) \neq \Theta(h_2)$).

Equivalently: Θ does not factor through the experimentally compressed state space $H/\sim_Y$.

Proof sketch. If the pair exists, no estimator on observable records can separate h_1 from h_2 . If no such pair exists, Θ is constant on each indistinguishability class and is identified. $\square$

This is the standard non-identification condition [Pearl, 2009] instantiated in the present typed setup.

Concrete check. Build a six-state system $\{a, b, c, d, e, f\}$. Let the observation function collapse $\{a, b\}$ to the same record. Set $\Theta(a) = 0$, $\Theta(b) = 1$. The kernel pair (a, b) fires: identical observation, different target. Residual ambiguity = $\log_2 2 = 1.000$ bit. No symmetry or automorphism is required. This is a straightforward indistinguishability argument, not a gauge-theory argument.

Where this falls short. Lemma 1 requires the compressed state space $H/\sim_Y$ to be a well-behaved mathematical object. On quantum systems, systems with non-abelian symmetries, or infinite-dimensional spaces, the quotient may not be constructible. Theorem 1 handles all of these.

5 The Descent-Obstruction Theorem

Lemma 1 catches the case where two states look identical and the target differs. There is a more general obstruction. Every *pair* of measurement contexts can be locally consistent while no single global assignment is compatible with all contexts simultaneously. This is contextuality, a real phenomenon in quantum mechanics that also applies to consciousness attribution. Theorem 1 states this local-to-global obstruction at the level of effective descent.

5.1 Insufficiency of Čech cohomology

The standard tool for contextuality is sheaf cohomology, developed by Abramsky and Brandenburger [2011] for quantum systems. A nonzero cohomology class $\check{H}^1(\mathcal{U}_Y, \hat{\Theta}) \neq 0$ certifies that local target assignments cannot be glued globally. This detects cases Lemma 1 does not.

$\check{H}^1$ is itself not the full story. It requires a good cover (geometric regularity), abelian target structure (addition and subtraction defined), and the obstruction in exactly degree 1. These conditions fail

for quantum POVM systems, for non-abelian symmetry groups, and for systems with non-standard topologies.

The kernel-pair case (Theorem 1) has $\check{H}^1 = 0$ but $D = 1$: a direct counterexample. A detector that checks only $\check{H}^1$ has recall = 1/2 on a minimal two-case suite. That is not good enough.

5.2 Effective descent

Imagine reconstructing the value of Θ by combining local measurements from overlapping contexts. *Effective descent* means there is a procedure that takes all the local results, checks overlap consistency, and assembles them into a single global answer. When descent is *not* effective, no such assembly exists. Every pair of local measurements can be consistent and the global assembly remains impossible.

Definition 8 (Descent-obstruction flag). *For admissible observation family Y' :*

$$D_{Y'}(X) = \begin{cases} 1 & \text{if } F|_{\mathcal{C}_{Y'}} \text{ has non-effective descent} \\ 0 & \text{otherwise.} \end{cases}$$

Definition 9 (Revision envelope). *Let Ω denote the declared physical, computational, and operational constraints on what structural changes to the observation/intervention framework are admissible. A structural revision r is any change to the admissible site that falls within Ω . $\text{Rev}_\Omega(Y)$ denotes the set of such revisions. A revision is conservative (Morita-equivalent) if it induces an equivalence of structural model categories over the same admissible site, preserving the target fiber, covering families, and context structure. A revision is non-conservative if it adds a genuinely new admissible observable, intervention, or target-relevant operation not definable from the old site.*

Definition 10 (Revision-stable persistent non-effective descent). *The identification obstruction is revision-stable if it survives every admissible structural revision:*

$$Z = f(X, Y) = \min_{r \in \text{Rev}_\Omega(Y)} D(r_*X) = 1.$$

$Z = 1$ means: no measurement protocol you try, no refinement, expansion, or structural revision of the observation family within Ω , allows the target to be globally assembled from local data. The prior formula $\min_{Y' \in \text{Adm}(Y)} D_{Y'}(X)$ is the special case in which Ω contains only refinements within a fixed structural language. Definition 10 extends this to all admissible structural revisions.

Proposition (Conservative Invariance). *If $e : \mathcal{L}_D \rightarrow \mathcal{L}_D^+$ is a Morita-conservative expansion over the admissible site (the induced functor on structural model categories is an equivalence compatible with the site topology and target fiber), then:*

$$F \text{ has effective descent in } \mathcal{L}_D \iff e_*F \text{ has effective descent in } \mathcal{L}_D^+.$$

Proof sketch. Conservative presentations induce equivalent descent categories only when the equivalence preserves the site topology: covering families, pullback compatibility, and the target fiber are transported together. Equivalences preserve existence and uniqueness of global objects. Effective descent is exactly existence of a global object realizing the local datum. No identification-failure premise is used. $\square$

Corollary. *Conservative expansion cannot overturn $Z = 1$. If the target fiber has non-effective descent, no Morita-equivalent re-presentation of the structural language changes this. Any future argument that a richer structural language supplies the sufficiency bridge must add a non-conservative admissible operation. That changes Y and must be declared in Ω rather than treated as a language-only clarification.*

5.3 The main criterion

Proposition (Finite deterministic factorization). *Let H and $\mathcal{R}_Y$ be finite sets, let $E_Y : H \rightarrow \mathcal{R}_Y$ be a deterministic observation map, and let $\Theta : H \rightarrow T$ be a target with decidable equality. An identification procedure using only admissible records exists if and only if there is a map $\hat{\Theta} : \mathcal{R}_Y \rightarrow T$ such that*

$$\Theta = \hat{\Theta} \circ E_Y.$$

Equivalently, Θ is constant on every fiber of E_Y .

Proof. If such $\hat{\Theta}$ exists, the procedure observes $r = E_Y(h)$ and returns $\hat{\Theta}(r)$. If an identification procedure using only records exists, its output is a function of r alone. Two hidden states in the same fiber of E_Y therefore receive the same output, so correctness requires Θ to be constant on that fiber. Define $\hat{\Theta}(r)$ to be that common value for each realized record. Unrealized records can be assigned arbitrarily. This is the finite deterministic form of effective descent: local record data assemble a unique global target value exactly when the target factors through the record object. $\square$

Proposition (Exact finite stochastic factorization). *Let H and $\mathcal{R}_Y$ be finite sets. Let $K : H \rightarrow \Delta(\mathcal{R}_Y)$ assign each hidden state an exact observation distribution over records, and let $\Theta : H \rightarrow T$ be a target with decidable equality. An exact identification procedure whose information is exhausted by the stochastic observation site exists if and only if Θ factors through the kernel map:*

$$\Theta = \hat{\Theta} \circ K$$

for some map $\hat{\Theta}$ defined on the image of K . Equivalently, whenever $K(h) = K(h')$, $\Theta(h) = \Theta(h')$.

Proof. If Θ factors through K , the procedure reads the exact kernel $K(h)$ and returns $\hat{\Theta}(K(h))$. Conversely, any exact procedure using only the stochastic site must assign the same answer to hidden states with the same observation distribution. Correctness therefore requires Θ to be constant on each fiber of K . Define $\hat{\Theta}$ on each realized kernel by that common value. $\square$

This proposition concerns exact distributional information. It does not say that one finite sample identifies the target. If two kernels overlap, one-sample classification can have nonzero Bayes error even when the exact kernels are distinct. The obstruction here is sharper: if the kernels are identical and the targets differ, no amount of sampling from that site identifies the target.

The claim that descent is the most general reconstruction principle is not asserted by fiat. Grothendieck’s theory [Grothendieck et al., 1971] establishes that effective descent over a site is precisely the condition under which coherent local data determines a global object. The propositions above prove the reduction for finite deterministic systems and exact finite stochastic systems. Any identification procedure (maximum-likelihood estimation, instrumental variables, Pearl’s do-calculus [Pearl, 2009], Bayesian updating) has the form: given local measurement records covering the state space at some observational resolution, recover the target parameter. That recovery has the structure of effective descent of the target fiber over the measurement site. The working claim is that when the target fiber has non-effective descent, no reconstruction procedure of this form can succeed, because the local data does not carry enough information to assemble a global answer. This reduction is argued here and proved for the formal components named in §13. The full proof that every admissible identification procedure in every substrate class factors through effective descent remains outside the scope of this article.

More precisely: for any site $(\mathbb{C}_Y, J)$, where J is a Grothendieck topology on the category of admissible measurement contexts, non-effective descent of the target fiber F over this site is the condition that no compatible family of local sections can be uniquely assembled into a global section. The same conceptual invariant (non-effective descent of the target fiber) applies across substrates, even though the computational tool used to detect it varies with the topology. For classical finite systems one uses Čech H^1 . For quantum POVM systems, the Abramsky–Brandenburger empirical-model method [Abramsky and Brandenburger, 2011]. For coalgebraic systems, bisimulation-invariant descent. The substrate-specific table below lists these implementations. The claim that each substrate instantiates the same descent criterion under its natural topology is a scoped mathematical program, not a completed proof inside this article.

Theorem 1 (Revision-stable persistent non-effective descent, scoped form). *For an experimentable theory (H, Y, E_Y, Θ) with declared revision envelope Ω , assume that admissible identification procedures for the substrate factor through effective descent of the target fiber over the observation site.*

Θ is necessarily non-identified from admissible observations if and only if $Z = 1$.

Proof sketch under the descent-factorization assumption. If $Z = 1$, then every admissible revision $r \in \text{Rev}_\Omega(Y)$ gives non-effective descent. No reconstruction of Θ from observable records exists for any admissible protocol or structural revision within Ω , since descent is the most general reconstruction principle and the Proposition (Conservative Invariance) rules out conservative expansion as a route around the obstruction. If $Z = 0$, some revision r gives effective descent, so the target fiber can be reconstructed from local data via the descent morphism. $\square$

Non-applicability boundary (formally stated). A substrate is in-domain (the theorem applies) if and only if all four of the following are defined: (1) an admissible context site $\mathbb{C}_Y$, (2) a target-fiber object $F \rightarrow H$, (3) descent semantics for that substrate category, and (4) a declared revision envelope $\text{Rev}_\Omega(Y)$. If any term is absent, the correct output is non-applicability (neither $Z = 0$ nor $Z = 1$) and the governance default is fail-closed.

5.4 The Sufficiency Frontier

Theorem 1 states when P1–P4 systems are in the non-identification class under the descent-factorization assumption. The question of whether every P1–P4 system is conscious (sufficiency direction) is addressed here as a formal boundary result, not left as a philosophical open question.

Conservative expansions cannot supply sufficiency. If a future structural language $\mathcal{L}_D^+$ is conservative over $\mathcal{L}_D$ (Morita-equivalent by the Proposition above), it adds no new structural distinctions between models:

$$\text{Mod}(\mathcal{L}_D^+) \simeq \text{Mod}(\mathcal{L}_D) \implies \Delta(\text{distinguishable structural cases}) = 0.$$

A conservative expansion cannot turn $\text{P1–P4} \not\Rightarrow \Theta$ into $\text{P1–P4} \Rightarrow \Theta$ unless the implication was already derivable from the old structure. No such derivation exists in the current framework. Conservative re-presentations of the language cannot supply the sufficiency bridge.

Non-conservative expansions change the domain. If $\mathcal{L}_D^+$ adds a primitive that directly tracks Θ , this is not a language clarification. It adds a new admissible probe, changing Y . The sufficiency bridge derived from $\mathcal{L}_D^+$ would then be a consequence of the new Y , not of the old structural framework. Any such expansion must be declared in Ω and evaluated as a non-conservative revision.

Conclusion. The negative sufficiency result (P1–P4 systems are not shown to be conscious) is not an artifact of an inadequate structural language. It is stable under all conservative expansions and explicitly defeasible only under declared non-conservative revisions that change what is admissible. Within any observable-relative framework, the bridge from the non-identification class to consciousness attribution requires an exogenous commitment that lies outside it. Other axiomatizations may add further commitments. The result here characterizes what observable-relative formal frameworks can determine without adding those commitments.

5.5 Comparison of approaches

Table 1 summarizes how the kernel-pair lemma, cohomological diagnostics, and the descent theorem cover the representative substrate classes used in the paper.

Case	Lemma 1	$\check{H}^1$	Thm. 1
Two states look identical, Θ differs	catches	misses	catches
Contextuality triangle (classical)	misses	catches	catches
Quantum POVM contextuality	n/a	catches	catches
Non-abelian symmetry system	partial	partial	catches
Identifiable control	silent	silent	silent

Table 1: Coverage across substrate classes. Lemma 1 is the H^0 special case of Theorem 1. $\check{H}^1$ is a sufficient diagnostic under classical good-cover conditions only.

Substrate-specific computation. In practice the right mathematical tool depends on the substrate:

Substrate	Use this obstruction tool
Classical finite systems	Čech H^1 ; CSP global-section check
Quantum (POVM) systems	Abramsky–Brandenburger empirical model [Abramsky and Brandenburger, 2011]
Non-abelian systems	Non-abelian H^1 , torsors, groupoid descent
Measurable/probabilistic	Invariant σ -algebras, measurable descent
Coalgebraic / process systems	Bisimulation-invariant descent

Table 2: Substrate-specific tools for computing the descent-obstruction flag.

Quantitative predictions. The checks are executable in Appendix A and the experiments directory.

- *Contextuality triangle:* δ^0 rank over $\mathbb{F}_2 = 2$; $\dim \check{H}^1 = 1$; zero global sections; $D = 1$.
- *Kernel-pair case:* $\check{H}^1 = 0$; $D = 1$; H^1 -only recall on two-case suite = $1/2$.
- *Identifiable control:* $D = 0$.

- *Finite stochastic kernel pair*: identical observation kernels with different targets fail exact factorization; distinct kernels pass exact factorization while retaining finite-sample error.
- *Boolean recurrent architecture*: report-only traces leave a latent target unidentified across all checked input sequences; a direct latent reader removes the obstruction by changing the site.

In the consciousness applications below, simpler branches do most of the work. IIT fails at the witness stage. GWT and higher-order theories generate kernel-pair worries when access or metacognitive evidence is used for a phenomenal target. The descent generalization matters for contextual, quantum-adjacent, and non-abelian substrates where those simpler diagnostics no longer cover the space of identification failures.

6 The Governance Corollary

If $Z = 1$ for a system, then no experiment (no certificate, audit, or disclosure protocol) can establish that Θ is identified. The mathematics establishes two things, which must be kept distinct.

The first is an epistemic claim that follows from Theorem 1 alone. Any governance protocol that issues a low-concern verdict on consciousness for such a system issues a verdict it cannot substantiate. The theorem shows no admissible experiment can supply the required identification evidence.

The second is a normative premise we adopt explicitly: a low-concern verdict requires epistemic support, and in the absence of any possible support the appropriate governance default is restriction rather than abstention. We adopt this premise on the grounds that the asymmetry of error costs under consciousness uncertainty warrants precautionary closure. The premise is not unconditional. It dominates when the downside of a mistaken low-concern verdict is severe, hard to reverse, and borne by a possible subject, while the cost of withholding the verdict is bounded by delay, further disclosure, narrower deployment, or a graduated precautionary regime. Where those conditions fail, the mathematics still gives the epistemic negative: the low-concern verdict lacks identification evidence. It does not by itself settle the practical policy response. Given this premise, the epistemic conclusion translates into the governance rule. The corollary is conditional: given the premise, restriction follows as a theorem. Without it, the mathematics establishes that low-concern verdicts are epistemically unwarranted, but does not specify the governance response.

Corollary 1 (Fail-closed verdict rule). *If $Z = f(X, Y) = 1$, then for every admissible $Y' \in \text{Adm}(Y)$:*

$$\Theta \notin \text{Desc}(Y') \implies \text{low-concern verdict on } \Theta \text{ is not admissible.}$$

Proof. By $Z = 1$ every Y' gives non-effective descent. A low-concern verdict requires establishing identifiability under some Y' . Non-effective descent entails no such demonstration exists. $\square$

The three-way verdict structure.

Situation	Correct verdict
Experimentability witness fails	Non-evaluable $\rightarrow$ no low-concern verdict
Witness holds and $Z = 1$	Identification impossible $\rightarrow$ verdict formally forbidden
Witness holds and $Z = 0$	Possible $\rightarrow$ requires adversarial certificate + replication

Both the “witness fails” and “ $Z = 1$ ” branches are fail-closed. A governance protocol open to low-concern verdicts in either branch is inconsistent with Corollary 1.

Many governance frameworks for AI consciousness are precautionary. They advise caution because the evidence is incomplete. The present corollary identifies a narrower case. When $Z = 1$ relative to a declared Ω , the declared evidence base cannot identify the target. Under the normative premise adopted above, a low-concern verdict lacks the support it requires.

Operational use. The corollary has a direct audit form. A low-concern verdict must state the site, target, revision envelope, classifier, challenge record, revocation rule, and provenance record on which it relies. If any required field is absent, the verdict is not evaluable under this criterion.

6.1 The Disclosure Requirement

The governance rule is operationalizable only if the system’s descent structure is declared. A low-concern claim that does not supply the following six elements is not evaluable and receives the same fail-closed verdict as $Z = 1$.

Any organization asserting a low-concern verdict for Θ must publish:

1. The context category/site C_Y : a complete specification of admissible measurements and interventions.
2. The target fiber object $F \rightarrow H$: a structural specification of Θ as a fiber over the hidden-state space.
3. The obstruction datum X : the descent data (transition maps, holonomy, kernel-pair constraints, contextual compatibility) that enables a third-party auditor to evaluate whether the system falls in the P1–P4 class.
4. The admissible revision envelope Ω : the declared physical, computational, and operational constraints on structural revision, with explicit classification of each proposed language expansion as conservative (Morita-equivalent) or non-conservative (genuinely new admissible probe or operation).
5. A revision-stable descent test result: a demonstration that $Z = \min_{r \in \text{Rev}_\Omega(Y)} D(r_*X) = 0$, with a certificate replicable by a third party under the declared Ω .
6. A non-applicability statement or confirmation: an explicit declaration that all four in-domain conditions (§5.3) are satisfied, or identification of which condition fails and why non-applicability rather than $Z = 0$ is the correct output.

Missing any element forces fail-closed. Missing Ω means Z is not evaluable under the revision-stable criterion, and undeclared non-conservative expansions permit post-hoc revision of the admissible site to avoid a $Z = 1$ verdict.

Each element addresses a specific governance failure mode. Undeclared site data permits gerrymandering the admissible evidence base. Undeclared holonomy data conceals whether the system falls in the P1–P4 class. Missing Ω permits strategic omission of non-conservative revisions that would resolve the obstruction. A missing revision-stable certificate allows unilateral identifiability claims that cannot be audited across structural change.

6.2 The Ω -Audit Protocol

The disclosure requirement names Ω as an input. It does not by itself make Ω hard to game. A claimant who wants a low-concern verdict can shape the envelope before the descent test is run,

and that choice can determine whether Z is evaluable. An Ω declaration therefore needs its own audit protocol.

The proposed protocol is organized around six failure modes. They are not assumed to be complete. They are a finite audit framework against which a stronger completeness theorem could later be tested.

Under-specification. A claimant may declare Ω vaguely, for example by saying that admissible revisions are those consistent with current physics or current practice. That lets later challengers be accepted or excluded after the fact. The remedy is finite enumeration. The declaration must list admissible revision classes: new observable types, intervention types, model-access regimes, target-relevant operations, and structural vocabulary changes. Each class must carry an inclusion or exclusion rationale. Anything outside the enumeration is inadmissible for the current certificate, but the enumeration itself remains open to public challenge.

Strategic exclusion. A claimant may exclude a revision class that would resolve the obstruction by calling it physically impossible, operationally infeasible, or out of scope. Exclusion is legitimate only if the reason is typed and auditable. The allowed reason types should be finite: physical impossibility, computational intractability, regulatory prohibition, or cost above a declared threshold. The declared threshold is part of the claim. If an excluded class lacks one of these reasons, or if the reason fails under audit, the low-concern verdict is revoked or reduced to non-applicability.

Misclassification. A claimant may add a probe that changes the admissible site while calling it a conservative language expansion. The audit rule is simple. Any expansion claimed as conservative must supply a witness that the new presentation preserves the target fiber, covering families, and context structure. In mathematical cases this is a Morita-equivalence witness. In applied cases it is the corresponding operational equivalence argument. No witness means the expansion is treated as non-conservative and triggers a fresh Z evaluation on the changed site.

Post-hoc revision. A claimant may alter Ω after seeing data or after receiving a challenge. The remedy is preregistration discipline. The envelope must be declared before certification, timestamped, and hashed. Any later change starts a new certificate rather than continuing the old one. The earlier verdict does not inherit evidence from the revised site unless the new evaluation is run.

Residual discovery failure. A claimant may satisfy the listed declaration rules while a material non-conservative probe class is never surfaced. Seeded omission tests do not solve this problem. They can show that a red team detects known planted defects under benchmark conditions, but they do not estimate the probability that an adversarially chosen future probe class remains outside the audit community’s search space. Seeded omissions should therefore be treated as a skill gate only. They cannot certify residual closure. For operational use, let ε_{esc} be the residual adversarial escape load and let ι_{audit} be verified audit integrity: independent selection, escrow funding, access parity, public challenge logs, conflict checks, and whistleblower channels. The admissibility score has the form

$$\sigma_{\text{adm}} = \iota_{\text{audit}}(1 - \varepsilon_{\text{esc}}).$$

If any required dossier field is absent, if access parity fails, if red-team independence is compromised, or if ε_{esc} cannot be bounded under the declared adversary model, the output is non-applicability. A seeded-omission success rate is not an admissible bound for ε_{esc} .

Apparatus provenance failure. A claimant may shape the people or infrastructure that generate the challenge record before the public record is fixed. The visible dossier can still show formal independence, escrow funding, access parity, public logs, and a timestamped envelope. The hidden

dependence sits upstream of those records. Relevant channels include funding, employment prospects, advisory ties, compute or API access, benchmark infrastructure, data access, publication access, shared investors, and informal reciprocal arrangements. For operational use, this residual is written as X_{prov} . A low-concern verdict therefore requires a provenance certificate for auditors, red teams, classifiers, panel members, infrastructure providers, and benchmark or data providers. The certificate must aggregate claimant-originated benefits by source class. Per-edge materiality thresholds may limit how much detail is published, but they cannot create a safe harbor against aggregation. If source-class provenance cannot be certified, the provenance gate fails and the output is non-applicability.

This protocol changes the burden of a low-concern verdict. The claimant must publish the envelope before the descent test, defend each excluded revision class, supply equivalence witnesses for conservative expansions, certify the provenance of the challenge apparatus, preserve challenger independence, and accept revocation when the envelope was too vague or moved after inspection. A third-party auditor need not settle consciousness. The auditor only asks whether the claimed low-concern verdict is supported by a fixed and contestable revision envelope.

Completeness problem. The six failure modes above are the main attacks this proposed protocol blocks. A later reduction theorem would show that every strategy for gaming an Ω declaration either leaves the envelope under-specified, strategically excludes a relevant revision class, misclassifies a site-changing operation as conservative, revises the envelope after evidence is observed, exploits a failure of the discovery apparatus, or contaminates the provenance of the apparatus that generates the public record. No such reduction theorem is claimed here. If such a reduction holds, the audit protocol is complete relative to the declared site and target fiber. If it fails, the counterexample identifies another failure mode and the protocol must be expanded. The hardest candidate counterexample is the future probe class that cannot be named at declaration time. A completeness proof must show that this reduces to one of the public-record failure modes, or else treat it as residual discovery failure. Operationally, this means that a low-concern dossier must not infer residual closure from planted omissions. It must either bound residual escape under a declared adversary model and a public challenge process, or fail closed. A second counterexample is provenance atomization: a claimant splits influence across sub-threshold proxy paths so that no single edge looks material while the aggregate claimant-originated exposure is material. The protocol blocks this only if the provenance certificate aggregates by source class. A five-mode reduction that places all such cases under residual discovery is too broad; it classifies by failed outcome rather than by a pre-specified mechanism. The natural formal target for follow-on work is a reduction theorem over these six public objects: declaration content, exclusion boundary, conservative classifier, certification timing, discovery apparatus, and apparatus provenance.

7 Worked Application: A Minimal Broadcast-Report Architecture

This section applies the criterion to a small cognitive architecture. The example is not offered as a model of biological consciousness. Its role is narrower: it shows how the site, target fiber, holonomy, and P4 condition are specified in a case where the obstruction can be checked by inspection.

Consider an architecture with three modules: a perceptual store P , a workspace W , and a report controller R . The hidden state $h \in H$ contains a latent episode token and three local encodings of that token, one in each module. The admissible observation site C_Y has three contexts:

$$U_{PW}, \quad U_{WR}, \quad U_{RP}.$$

Each context exposes pairwise agreement records between two modules. No admissible probe reads all three encodings at once. This restriction represents an ordinary instrumentation limit: the available protocols can test pairwise access, workspace availability, and report alignment, but cannot jointly identify the latent episode token across all modules in a single record.

The target Θ is global episode identity: whether the three local encodings are assignments of one and the same latent episode. The target fiber $F \rightarrow H$ carries a local binary label for the episode token in each context. On overlaps, local labels are related by transition maps. Take two transition maps to preserve the label and one to flip it:

$$g_{PW,WR} = +1, \quad g_{WR,RP} = +1, \quad g_{RP,PW} = -1.$$

The product around the cycle is -1 . Every pairwise comparison is locally satisfiable, but no global label can be chosen consistently across all three contexts. The target fiber therefore has non-effective descent over C_Y . The finite check is executable in Online Resource 1.

The P1–P4 check is now explicit. P1 holds because each pairwise context supports a local assignment of the episode label. P2 holds because overlap assignments are related by transition maps, and the cycle product prevents global compatibility. P3 holds because the induced context-groupoid action has no global fixed section. Equivalently, transporting a local label around the cycle returns its opposite. P4 holds relative to the following declared revision envelope: admissible revisions may add time resolution, duplicate pairwise probes, replace labels by conservative aliases, or add pairwise behavioral readouts, but may not add a direct all-context token reader. Each admissible revision commutes with the transition maps and preserves the cycle product -1 . Thus no revision inside Ω dissolves the obstruction.

The worked example also shows what would count as a real defeat of the obstruction. A probe that reads the latent episode token jointly across P , W , and R would change the site. It is a non-conservative revision and must be declared in Ω . If such a probe is physically and operationally admissible, the case moves from $Z = 1$ to $Z = 0$ or to a new site requiring a fresh computation. If it is not admissible, the low-concern verdict remains unsupported.

This example matters for the larger paper in two ways. First, it shows that P1–P4 are not merely labels attached after the fact. They are a checklist: context site, target fiber, transition maps, holonomy, and revision envelope. Second, it separates access from phenomenal attribution. The architecture can exhibit pairwise broadcast and report alignment while still failing to identify the global target. That is the form of the test the later theory sections apply to IIT, GWT, and HOT at a higher level of abstraction.

A second executable check uses a small Boolean recurrent architecture rather than the pairwise-context cycle. The system has an input bit, an observable report bit, a memory bit with update rule $m_{t+1} = x_t$, and a latent bit with update rule $a_{t+1} = a_t$. The report-only site sees identical traces for two states that differ only in the latent bit, across every input sequence up to the checked horizon. A direct latent reader resolves the target, but only by changing the site. The check is supplied in Online Resource 1 as `boolean_recurrent_obstruction_check.py`.

8 Worked Disclosure Dossier: Llama 3.1 8B-Instruct

This dossier is a positive control for the framework, not an empirical claim about LLM consciousness. It shows that ordinary access/report evidence can be accepted for a narrow target while refusing to

transfer that evidence to phenomenal consciousness. The section applies the disclosure requirement to a public system rather than a toy architecture. The system is Meta Llama 3.1 8B-Instruct, an open-weight, text-only, instruction-tuned transformer released in July 2024 [Grattafiori et al., 2024, Meta, 2024]. The model card describes it as an autoregressive language model using an optimized transformer architecture, with supervised fine-tuning and RLHF for helpfulness and safety alignment. For the 8B model, the public record states a 128k context length, grouped-query attention, multilingual text input, text and code output, pretraining on more than 15 trillion tokens, and a December 2023 pretraining data cutoff.

The dossier below is deliberately narrow. It concerns one self-hosted inference instance of the released 8B instruction model, without external tools, memory, autonomous action loops, fine-tuning, or agent scaffolding. Those additions would change the site and require a fresh dossier.

The access/report probe in this section is not the paper’s novelty. Mechanistic interpretability already supplies many ways to test whether representations in a transformer track prompt content and affect later output. The point of the dossier is the audit distinction. Evidence that the model uses a context item is evidence for an access/report target. It is not, by itself, evidence for a phenomenal target. The framework’s contribution is to keep those targets separate when a verdict is issued.

1. Observation site. The admissible site contains: public model-card information; released model weights and architecture; token-level input-output behavior; prompt interventions; logit traces; activation probes; ablations of attention heads, MLP blocks, layers, residual streams, and safety wrappers; and reproducible self-hosted inference logs. It excludes non-public training traces, vendor-internal human-feedback records, private safety-classifier decisions, and unlogged deployment-layer state. It also excludes biological measurements, because the system is not biological.

2. Target fiber. Two targets must be kept separate. For access or report, the target fiber contains operational labels such as “information is available for token generation,” “the model reports a state,” or “the model can use a context item in later output.” Those labels can be tested with ordinary behavioral and mechanistic probes. For phenomenal consciousness or moral patienthood, the target fiber contains a phenomenal-status label for an inference state. The public site supplies no decidable equality test for that label.

3. Revision envelope. The declared Ω permits longer prompt batteries, adversarial prompts, activation patching, sparse-feature probes, causal ablations, alternative decoding parameters, quantization checks, and removal or replacement of deployment guardrails. It also permits comparison across self-hosted replicas of the same released weights. It does not permit adding persistent memory, tool use, online learning, multi-agent scaffolding, biological sensors, or private training data while calling the result the same site. Those are non-conservative changes and require a new Z evaluation.

4. Classifier. For access/report targets, the classifier is behavioral and mechanistic: does a probed representation causally affect later token generation or report behavior under held-out prompts? For phenomenal consciousness, no accepted classifier is supplied by the public record. Self-reports about consciousness are outputs under the same observation site, not labels for the hidden target.

5. Challenge record. A sufficient public challenge record would include prompt suites, activation probe protocols, ablation results, negative controls, model/version hashes, sampling parameters, deployment-wrapper descriptions, and failed attempts to find a non-conservative probe that changes the verdict. The current public record contains useful model-card and benchmark information, but it is not a consciousness-attribution challenge log.

6. Revocation rule. The verdict must be recomputed if the site changes. Examples include tool integration, persistent memory, online adaptation, fine-tuning, changed safety wrappers, multimodal inputs, autonomous action loops, or access to private training traces. The verdict also changes if a future theory supplies a decidable phenomenal classifier over this class of systems.

7. Provenance record. For a low-concern certificate, the audit would need to name who selected the probe classes, who ran the red-team process, who maintained the classifier, which infrastructure was used, and whether those actors or facilities had claimant-originated dependencies. The provenance record here is intentionally limited: the probe packet was selected and run by the author, the GPU infrastructure was rented, the model came from an ungated public mirror after the official gated repository was inaccessible, and no independent red-team or source-class provenance certificate was produced. This is sufficient for a worked access/report illustration. It is not sufficient for a low-concern phenomenal-consciousness certificate.

Verdict. For the access/report task in this dossier, the result is evaluable and yields $Z = 0$: the public model can be probed, ablated, and compared across replicates for that target. For phenomenal consciousness or moral patienthood, the dossier is non-applicable. The reason is not that Llama 3.1 8B-Instruct is shown to be non-conscious. The reason is that the public site lacks a target fiber with decidable equality, lacks a phenomenal classifier, lacks a public consciousness challenge log, and lacks the provenance certificate needed for a low-concern verdict. The framework therefore blocks a certified low-concern claim for the phenomenal target under this dossier, while allowing ordinary scientific claims about access, report, and mechanism.

This result is also a check on the framework’s own scope. The dossier does not infer consciousness from opacity. It separates targets. It accepts the system as experimentally tractable for access and report. It refuses only the stronger phenomenal verdict, because the dossier does not contain the objects that such a verdict requires. The executable access/report probe packet is `llama31_activation_probe.py` in Online Resource 1. It freezes the prompt schema and declared site in a manifest, then on a GPU collects selected hidden states, reports leave-one-out probe accuracy by layer, and runs a targeted residual-direction intervention. The packet is evidence about access/report evaluability only. On 2026-05-06, the packet was run on an A10 GPU against the ungated `NousResearch/Meta-Llama-3.1-8B-Instruct` mirror. The run used 24 controlled key-value prompts. Token-level answer accuracy was 24/24. Mean correct-label logit margin was 6.107, while the no-mapping control margin was approximately zero (-0.026). Leave-one-out centroid probes on selected hidden states reached 1.00 accuracy at layers 16, 24, and 31, and 0.75 at layer 8. A targeted residual-direction intervention reduced the correct-label margin at layers 16, 24, and 31, with mean deltas -0.732 , -1.669 , and -1.107 . These results support the narrow access/report claim under the declared site. They do not supply a phenomenal-consciousness classifier. The saved packet is `llama31_probe_packet_nous/summary.json` in Online Resource 1.

This experiment should be read as a positive control, not as a new mechanistic-interpretability result. Linear probes and residual interventions are familiar tools for showing that transformer representations track prompt content and affect later output. The point here is different: the framework accepts that ordinary evidence for the access/report target, assigns it the expected evaluable verdict, and then refuses to transfer that verdict to the phenomenal target without the missing target fiber, classifier, challenge record, revision envelope, and provenance certificate. If the access/report control had failed, the dossier would have been uninformative even for the narrow target. Because it succeeds, it shows that the non-applicability verdict for the phenomenal target is not produced by a blanket skepticism about empirical AI systems.

9 Implications for IIT, GWT, and Higher-Order Theories

The three cases below do different jobs. IIT is the hardest case for the experimentability witness, because its central quantity is defined at a level that current measurement does not reach. GWT and higher-order theories are cleaner as theories of access and metacognition. Their difficulty appears when those narrower targets are used as evidence for phenomenal consciousness.

9.1 Integrated Information Theory

IIT identifies consciousness with integrated information Φ above a threshold [Tononi et al., 2016]. For the present paper, the important point is not whether IIT is true. The important point is what would have to be measured for a full Φ -based identification claim to be auditable. Φ requires the cause-effect repertoire of every possible subsystem, across all spatial and temporal scales. For a system of N binary neurons this requires 2^N perturbation experiments. No current neuroimaging or electrophysiology protocol comes close. The observation record $\mathcal{R}_Y$ under any known experimental protocol does not contain the information needed to compute Φ with decidable equality.

IIT’s target Θ falls in the prior class for the purpose of full Φ -based identification, because the observation protocol required to test the target does not exist. It is therefore not yet in the “ $Z = 1$, identification impossible” class. The gap is specific. Full cause-effect repertoires are not in $\text{Adm}(Y)$ under any currently realizable experimental protocol.

A clarification is needed. IIT’s *predictions* are falsifiable, and one salient preregistered prediction was challenged in the 2025 adversarial study [Cogitate Consortium et al., 2025] (Cogitate consortium, $n = 256$, multimodal fMRI/MEG/iEEG): the expected sustained posterior synchronization was not found. The same study also challenged GNWT, especially through the lack of ignition at stimulus offset and limited prefrontal representation of some conscious dimensions. Those results are scientifically relevant, but they do not settle the identification question used here. The witness-failure verdict applies to a different question: whether the value of Φ can be computed from experimental data. Falsifying theory-derived predictions does not require computing Φ ; it requires only that the theory entails observable consequences. The identification problem arises when no observation protocol can supply the quantity the theory defines as the target (Φ itself). Current approximation schemes such as Φ^* , MIB, and GeoMIP are best read as proxies or restricted computations, not decidable equality tests for the theoretical target. The best implementations handle small systems far below brain scale, and none cited here claims decidable equality of $\hat{\Phi}$ with Φ for the full target.

Addressing the tractability objection. IIT proponents may argue that Φ is in principle computable from complete cause-effect structure, with the computation being exponential in N but tractable in principle. The experimentability witness requires *decidable equality*: a procedure that definitively answers “does this measured quantity equal the theory-defined target Φ ?” This is a different demand from tractability. The witness requires both (a) an observation protocol Y whose record $\mathcal{R}_Y$ contains the cause-effect repertoire, and (b) a decidable equality check between the measured quantity and the theoretical Φ . The exponential-computation barrier blocks (a) in practice. No current approximation scheme satisfies (b). Each Φ^* variant is an acknowledged lower bound or proxy; none constitutes a decidable equality test against the theoretical target. The witness-failure verdict rests on the absence of a decidable specification. Hanson and Walker [2019] independently established a related substrate-sensitivity problem for IIT: Φ is not invariant under state-labeling isomorphisms, so functionally identical systems can receive different Φ values. The descent-obstruction analysis here gives a substrate-neutral structural reason for that failure of invariance. IIT lacks the experimentability witness, meaning decidable equality on Θ , required for a full Φ -based identification claim. A version

of IIT that specified a practical approximation $\hat{\Phi}$ with a decidable equality test would clear this first hurdle. Whether $\hat{\Phi}$ actually tracks Φ would then become the next empirical question.

9.2 Global Workspace Theory

GWT identifies consciousness with global broadcast: information made available to multiple cognitive subsystems at once [Baars, 1988, Dehaene and Changeux, 2011]. This is a much more tractable target than Φ . Broadcast can be tested with masking, report, attention, and neural-imaging paradigms. For access consciousness, in Block’s sense [Block, 1995], GWT can supply the experimentability witness. This does not mean current GNWT predictions are unproblematic. The Cogitate adversarial study challenged GNWT’s offset-ignition and prefrontal-content predictions. The point here is narrower: broadcast/access targets can be specified in an experimentable record type even when particular GNWT predictions fail.

The trouble begins when broadcast evidence is used to license claims about phenomenal consciousness. Two systems with identical global broadcast patterns may differ in phenomenal experience. If the governance target is phenomenal consciousness, the kernel pair fires: (broadcast + phenomenal experience) versus (broadcast + no phenomenal experience) produce identical records under any broadcast-based measurement. GWT evidence can be strong evidence about access while still failing to settle the phenomenal target.

9.3 Higher-Order Theories

Higher-order theories identify consciousness with a representation of being in a mental state [Rosenthal, 2005, Carruthers, 2000]. Here again the first step is feasible. Metacognitive reports, confidence judgments, and prefrontal-monitoring signals can be placed in a record type with decidable equality. The harder question is whether higher-order representation identifies phenomenality. Blindsight cases make the problem vivid: subjects can show information-guided behavior in the absence of ordinary visual experience. If the kernel pair (higher-order representation + phenomenal experience) versus (higher-order representation + no phenomenal experience) is empirically realizable, Lemma 1 applies. HOT therefore passes the witness for a metacognitive target. It does not by that fact alone identify phenomenal consciousness.

10 Application: The Upload Problem and Evolutionary Selection

The upload problem. A digital system D is proposed as conscious by copying the behavioral profile of biological system B : $O(B) \approx O(D)$. Observable copying transfers the record $E_Y(h)$, not the descent structure (transition maps and holonomy of the target fiber). A verdict requires demonstrating $(C_{Y_D}, F_D, \rho_D) \cong (C_Y, F, \rho)$. Behavioral matching alone is insufficient; without structural isomorphism the correct verdict for D is non-evaluable.

The evolutionary argument. Natural selection acts on observable outputs, not on Θ directly. If Θ is descent-obstructed, selection on O does not imply selection on Θ . The inference “consciousness evolved because it confers fitness advantage” is structurally invalid when $Z = 1$: selection may have favored hidden causal structures X that produce adaptive O , with the obstructed property Θ as a byproduct.

Substrate-neutral identification. A digital system that instantiates an equivalent context category, transition maps, and holonomy structure, $(C_{Y_D}, F_D, \rho_D) \cong (C_Y, F, \rho)$, falls in the same

identification class as a biological system with the same structure. The relevant comparison is descent equivalence across the declared sites. The theory predicts that substrate-isomorphic systems are identification-equivalent when the site, target fiber, and revision envelope match.

Information flow is not holonomy. A system maximizing integrated information, global broadcast, or any other information-theoretic quantity satisfies P1 (local contextual presentability in information terms) but need not satisfy P3 (nontrivial holonomy). Information-theoretic magnitude is measurable and identifiable. Structural holonomy over the admissible observation site is what generates the obstruction. Large information flow is consistent with $Z = 0$.

11 Relation to the Literature

The mathematics in this paper is not new. Kernel-pair identification is standard in causal inference [Pearl, 2009]. Čech cohomology for contextuality is Abramsky and Brandenburger [2011]. Grothendieck descent is 60-year-old algebraic geometry [Grothendieck et al., 1971]. We apply known mathematical machinery to a new application domain.

The philosophical setting is older and less tidy. Nagel [1974] and Levine [1983] made the epistemic difficulty of subjective experience hard to ignore. Chalmers [1995] gave that difficulty its contemporary name. Dennett [1991] and Frankish [2016] press in the opposite direction: the work may be to explain the judgments, reports, and seeming of phenomenality without positing a further hidden property. This paper does not decide that dispute. It asks a narrower question: given a declared target, can the target be identified from the admitted observations? Block [1995] matters here because the access/phenomenal distinction is exactly where many proposed AI tests quietly change targets.

The stance of this paper is methodological rather than metaphysical. If the target is access, report, metacognitive monitoring, or functional integration, then ordinary experimental and mechanistic tests may identify it. If the target is phenomenal consciousness in the sense at issue for Nagel [1974], Levine [1983], and Chalmers [1995], the claimant must say how that target enters the observation site. An illusionist or behaviorist may instead deny that an additional phenomenal target is needed. That is a legitimate change of target. The present framework then evaluates the replacement target, not the discarded one. This is why the result does not settle the dispute between realist and deflationary theories of consciousness. It gives a discipline for moving between them: state the target, state the site, and show whether the target is identified.

More precisely, the paper adopts epistemic structuralism about scientific attribution, not metaphysical structuralism about what exists. Against the realist, it does not deny that there may be a fact of the matter beyond the declared observations. It says that such a fact is not licensed as a scientific verdict by that observation site. Against the illusionist or behaviorist, it does not require a further phenomenal residue. If the proposed target is report, judgment, seeming, or illusion-detection, the framework evaluates that target instead. Against a careless use of Block’s distinction, it insists that access evidence can be valid evidence for access while still failing to identify phenomenal consciousness. The framework is neutral about which target one should care about, but not neutral about target changes. A target change has to be declared.

The philosophical contribution is the identification reframing: a consciousness verdict must keep target, site, and admissible revision envelope fixed before treating evidence as decisive. Lemma 1 already gives a strong version of that point for classical indistinguishability cases, and Block’s access/phenomenal distinction explains why access evidence cannot simply be moved to the phenom-

enal target. The descent machinery generalizes the point to contextual and non-classical substrates where kernel-pair quotients or Čech H^1 are too narrow. Its fully general reduction remains a stated obligation; the finite factorization cases and the conservative-invariance results establish the part used here. From this we derive a governance corollary with an explicit six-element disclosure requirement that includes the revision envelope Ω , and we evaluate major consciousness frameworks against the experimentability precondition. **On formal frameworks for AI consciousness.** Butlin et al. [2023] set the current methodological baseline: derive computational indicator properties from GWT, IIT, HOT, and related theories, then assess AI systems against them. The approach is carefully empirical, but it presupposes that the indicator properties are experimentally accessible. Phillips and Tsuchiya [2024] show that all six IIT axioms can be restated as universal mapping properties in the sense of category theory, including the sheaf condition. Our paper extends their framework from definition to testability: the sheaf condition’s failure (absence of global sections) is exactly the descent-obstruction criterion we use as an identification-impossibility certificate. Kleiner and Tull [2021] give a categorical formulation of IIT; their goal is to define consciousness, ours is to characterize when such definitions are scientifically evaluable. Gunji et al. [2026] give a recent categorical reconstruction of contextuality in which orthomodular structure arises from classical Boolean contexts. That work is adjacent to our use of contextual obstruction, while our target is identification of a hidden property from admissible observations. Campero et al. [2025] provide the clearest taxonomy of impossibility-class arguments against AI consciousness; our result contributes a new category: identification-level impossibility independent of substrate implementation. Phua [2025] use artificial agents and ablations to test functional predictions associated with GWT, IIT-adjacent markers, and HOT. Their dissociations support the need to keep access, integration, and metacognitive targets separate before making any phenomenal-consciousness verdict.

Closest adjacent prior work. Inoué [2026] develops a sheaf-cohomological treatment of neural integration on brain manifolds, modeling local cognitive functions as sheaf sections and classifying integration failures as cohomological obstructions. Our framework differs in two specific ways. First, we use Grothendieck effective descent rather than Čech H^1 as the obstruction criterion; the kernel-pair example (Lemma 1) shows that descent is strictly more general. Systems exist with $H^1 = 0$ but $D = 1$, so an H^1 -only diagnostic misses part of the identification-failure class. Second, our target is the identification problem (whether the property Θ is recoverable from admissible measurements), not neural integration. The Proposition (Conservative Invariance) and the revision envelope Ω address the language-stability of the identification claim, which is not the question Inoué addresses.

On governance under consciousness uncertainty. Long et al. [2024] and Butlin and Lappas [2025] argue that precautionary governance is warranted. Our governance corollary sharpens their recommendation: for the hidden-variable substrate class, the uncertainty is identification-theoretic under the declared observation regime. On the normative premise adopted here, a low confidence score of “likely not conscious” is not enough. Wolfson [2026] and Rost [2026] independently anticipate related protection and readiness concerns through non-mathematical reasoning. Birch [2024] develops a graduated precautionary framework for sentience attribution; Corollary 1 gives a formal condition under which the evidential basis for low concern is absent. The framework in Zhou et al. [2025] defaults to “presumption of no consciousness”; the present theorem rejects that default when descent obstruction holds and the governance target requires identification evidence. The absence of admissible evidence does not license a negative verdict under non-effectiveness.

12 Objections

The revision-envelope objection. A critic may say that Ω is a knob. If Ω is too narrow, the obstruction is protected from serious future tests. If Ω is too broad, the result becomes hard to evaluate. This is the main practical pressure point of the framework. The answer is procedural: Z is always relative to a declared Ω , and the declaration is part of the public claim. A low-concern verdict therefore carries an audit burden. The claimant must say which new observables, interventions, probes, and structural vocabularies are admissible, and why excluded revisions are outside the physical or operational scope. An under-specified Ω does not produce $Z = 1$ or $Z = 0$. It produces non-applicability and the same fail-closed governance result.

The behaviorist objection. If two systems are indistinguishable under every admissible observation, a behaviorist may deny that any remaining difference is scientifically meaningful. The paper accepts the force of that objection at the epistemic level. When the target is not identified, science cannot license a positive or negative target attribution from the declared observations. The governance conclusion adds a further premise: a low-concern verdict is a decision with moral stakes, and such a decision requires identification evidence. The mathematics blocks the evidential claim. The normative premise determines the default response.

The substrate objection. The framework is substrate-neutral only at the level of form. Substrate still matters through the declared inputs. The admissible site Y , the target fiber $F \rightarrow H$, and the revision envelope Ω must be specified for each substrate. A biological system, a digital system, and a quantum-contextual system may instantiate the same descent pattern, but that equivalence must be shown. Substrate enters through the site and the allowed revisions. The theorem says what follows once those inputs are fixed. It does not remove the burden of fixing them.

13 Scope and Limitations

Not claimed: which systems are conscious. The theorems characterize when science cannot answer this question.

Not claimed: that $Z = 1$ implies consciousness is absent. It implies the question cannot be settled by admissible experiments.

Not claimed: that IIT, GWT, and HOT are false. The §9 evaluation is structural, not a truth judgment.

Not claimed: that the governance corollary covers all moral-status decisions.

Not claimed: that proof sketches constitute proofs. The finite deterministic and exact finite stochastic factorization results are proved in §5.3. Two broader reductions remain argued rather than proved: (1) that every admissible identification procedure across all substrate classes factors through descent, and (2) that the descent criterion is computed identically across all five substrate classes in Table 2. The paper’s contribution is to prove the finite components used below, isolate the broader central claims, and state the remaining proof obligations explicitly.

Formally verified components. Four claims in this paper have been mechanically verified in Lean 4 (Mathlib v4.30.0-rc2): the finite kernel-pair non-factorization lemma, the finite-alias Conservative Invariance case, a categorical sheaf-transport statement under suitable equivalence, and the Appendix A Case 4 Z -formula evaluations. The Lean files are supplied as Online Resource 2 and compile with only the standard axiom dependencies. They are verification receipts for named

components, not a proof of the full descent-factorization reduction. That general claim remains argued, not proved, and is the main central reduction still open.

Principal remaining gap: Ω operationalization. The revision envelope Ω must be declared and audited by a claimant before Z is evaluable. For real physical and computational systems, specifying and publicly justifying Ω is open-ended. Practitioners may under-specify, strategically exclude plausible non-conservative revisions, or dispute which expansions count as conservative. This does not affect the theoretical result (Z is well-defined relative to any declared Ω) but it places the evidential burden on Ω 's completeness and public auditability. The practical next step is therefore an Ω audit protocol: a declaration schema, an adversarial challenge process for missing revisions, and a revocation rule when a low-concern verdict rests on an under-specified or strategically narrow envelope. Such a protocol must also refuse a common shortcut: success on seeded omissions qualifies challenger competence but does not bound the probability that adversarial probe classes remain unsurfaced. Residual escape must be bounded under a declared adversary model and public challenge process, or the dossier is non-applicable. The same point applies to independence. Absence of a visible conflict is not a provenance certificate. The dossier must certify the source-class provenance of auditors, red teams, classifiers, panel members, infrastructure providers, and benchmark or data providers. Claimant-originated benefits must be aggregated across proxy and sub-threshold paths. Without that protocol, the framework can state non-applicability cleanly, but a real disclosure dossier remains too easy to game.

Completed checks. Internal adversarial review should be treated as a source of candidate objections only. It does not certify the theorem. Evidence for the framework has to come from public proofs, public code, worked systems, or external audit records. Four immediate checks have now been completed: §8 gives a public-system dossier for an open-weight transformer and a reproducible activation-probe packet for the access/report target; the finite stochastic script checks the exact kernel case; the Boolean recurrent script runs the obstruction calculation on a small transition system; and the prediction packet states the following falsifiable disclosure-packet prediction: for a fixed text-only, self-hosted, open-weight transformer with no external tools, memory, online learning, or autonomous action loop, access/report use of an input item should be evaluable from public weights, token-level behavior, logits, hidden states, activation probes, causal ablations, prompt interventions, and reproducible inference logs; phenomenal consciousness or moral patienthood should remain non-applicable under that same site unless the claimant supplies a target fiber with decidable equality, a public classifier, a public challenge record, a declared revision envelope, and a provenance certificate for the challenge apparatus. The prediction fails if an auditor supplies those phenomenal-target objects without adding a non-conservative probe, or if the access/report packet cannot produce above-chance behavioral evidence for a system known to use the relevant input.

Future work. First, extend the public-system dossier to a deployed system with tool use, memory, and logged deployment state. Second, extend the finite proof from exact kernels to finite-sample statistical decision procedures and finite measurable systems before claiming the fully general version. Third, submit the prediction packet to an external registry or independent auditor before using it as external evidence. Fourth, continue the philosophy-of-mind engagement in §11 by stating exactly which target is at issue: access, report, integration, metacognition, or phenomenal attribution.

14 Conclusion

For some hidden-variable systems, consciousness attribution has an identification problem that cannot be removed by additional observations inside the declared site. The case occurs when

the target fiber has revision-stable non-effective descent. Section 3 characterizes the systems in this class by four structural properties (P1–P4) that generate necessary non-identification for any substrate that supplies the required site, target fiber, descent semantics, and revision envelope. The revision-stable descent criterion (Theorem 1) states the impossibility under the descent-factorization assumption, and the Proposition (Conservative Invariance) shows that conservative re-presentations of the structural language do not remove the obstruction. Section 5.4 identifies the boundary of the framework: the bridge from P1–P4 to consciousness attribution requires an exogenous commitment or a non-conservative revision of the admissible site. The governance corollary derives from these. When $Z = 1$, a low-concern verdict lacks the identification evidence it requires. Any organization issuing such a verdict without first satisfying the six-element disclosure requirement, including a declared revision envelope Ω , has not supplied the evidence its verdict needs.

The worked application in §7 is intentionally small. That is the point of it. The framework becomes usable only when the claimant writes down the observation site, target fiber, transition data, and revision envelope for the system at hand. Without those objects on the page, the consciousness verdict is premature.

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A Falsification Suite

All quantitative predictions are confirmed by the following code, runnable with no dependencies beyond Python’s standard library. The suite tests four claims: (1) the contextuality triangle has $D = 1$ and $\check{H}^1 = 1$; (2) the kernel-pair case has $D = 1$ and $\check{H}^1 = 0$, demonstrating $\check{H}^1$ -only recall = 0.5; (3) the identifiable control has $D = 0$; and (4) the revision-stability predictions, namely that a conservative language expansion leaves D unchanged and a non-conservative hidden-state-revealing revision removes the obstruction.

```
import itertools

def rank_f2(matrix):
    m = [row[:] for row in matrix]
    if not m:
        return 0
    rows, cols = len(m), len(m[0])
    rank = 0
    for col in range(cols):
        pivot = next(
            (r for r in range(rank, rows) if m[r][col] % 2 == 1), None)
        if pivot is None:
            continue
        m[rank], m[pivot] = m[pivot], m[rank]
        for r in range(rows):
            if r != rank and m[r][col] % 2 == 1:
                m[r] = [(m[r][c] ^ m[rank][c]) for c in range(cols)]
        rank += 1
    return rank

# Case 1: contextuality triangle
CONTEXTS = ["C12", "C23", "C13"]
CONTEXT_STATES = {
    "C12": ("h1", "h2"), "C23": ("h2", "h3"), "C13": ("h1", "h3")}

def overlapping_states(ca, cb):
    return set(CONTEXT_STATES[ca]) & set(CONTEXT_STATES[cb])
```

```

def local_value(ctx, state, section):
    return section[CONTEXT_STATES[ctx].index(state)]

def is_globally_consistent(sections):
    for ca, cb in itertools.combinations(CONTEXTS, 2):
        for state in overlapping_states(ca, cb):
            if local_value(ca, state, sections[ca]) != \
               local_value(cb, state, sections[cb]):
                return False
    return True

ODD_PARITY = {
    ctx: [(a,b) for a in (0,1) for b in (0,1) if (a^b)==1]
    for ctx in CONTEXTS}

def global_sections(local_family):
    return [
        {ctx: sec for ctx, sec in zip(CONTEXTS, combo)}
        for combo in itertools.product(
            *[local_family[c] for c in CONTEXTS])
        if is_globally_consistent(
            {ctx: sec for ctx, sec in zip(CONTEXTS, combo)})]

coboundary      = [[1,1,0],[0,1,1],[1,0,1]]
triangle_rank   = rank_f2(coboundary)           # 2
triangle_h1     = 3 - triangle_rank             # 1
triangle_gs     = len(global_sections(ODD_PARITY)) # 0
triangle_D      = 1 if triangle_gs == 0 else 0   # 1

assert triangle_rank == 2
assert triangle_h1   == 1
assert triangle_gs   == 0
assert triangle_D    == 1

# Case 2: kernel-pair ( $H^1 = 0$ ,  $D = 1$ )
obs      = {"a": "same", "b": "same"}
target   = {"a": 0, "b": 1}

def kernel_pair_D(obs, theta):
    for x, y in itertools.combinations(obs, 2):
        if obs[x] == obs[y] and theta[x] != theta[y]:
            return 1
    return 0

kernel_h1 = 0
kernel_D  = kernel_pair_D(obs, target)    # 1
assert kernel_h1 == 0
assert kernel_D  == 1

# Case 3: identifiable control
obs_ctrl  = {"a": "obs_a", "b": "obs_b"}
tgt_ctrl  = {"a": 0, "b": 1}
control_D = kernel_pair_D(obs_ctrl, tgt_ctrl) # 0
assert control_D == 0

# Detector comparison
total      = triangle_D + kernel_D
h1_detected = (1 if triangle_h1>0 else 0) + (1 if kernel_h1>0 else 0)

```

```

h1_recall      = h1_detected / total      # 0.5
descent_recall = total / total            # 1.0

assert h1_recall      == 0.5
assert descent_recall == 1.0

```

Note on Case 4. This case is a definitional verification, not an independent empirical test. The Conservative Invariance proposition guarantees by category-theoretic argument that adding a Morita-equivalent alias preserves the descent flag. The non-conservative variant adds an observable that distinguishes the two hidden states, which by the definition of the descent obstruction must resolve it. Case 4 is included as a verification benchmark to confirm the formal machinery behaves as the proposition predicts. It does not constitute evidence beyond what the proposition's category-theoretic content already establishes.

```

# Case 4: revision-stability benchmark
# Base: kernel-pair obstruction (D_base = 1).
# Conservative revision: add a definitional alias of the existing observable.
#   The Proposition (Conservative Invariance) predicts D unchanged.
# Non-conservative revision: add an observable that distinguishes the hidden
#   states.
#   This changes Y; the obstruction is resolved (D = 0).

def compute_D(obs_map, theta_map):
    for x, y in itertools.combinations(obs_map, 2):
        if obs_map[x] == obs_map[y] and theta_map[x] != theta_map[y]:
            return 1
    return 0

obs_base      = {"a": "obs_X", "b": "obs_X"}
theta_base    = {"a": 0, "b": 1}
D_base        = compute_D(obs_base, theta_base)
assert D_base == 1, "Base kernel-pair must have obstruction"

obs_conservative = {"a": ("obs_X", "obs_X_alias"), "b": ("obs_X", "obs_X_alias")}
theta_conservative = theta_base
D_conservative   = compute_D(obs_conservative, theta_conservative)
assert D_conservative == 1, "Conservative expansion must preserve obstruction"
assert D_conservative - D_base == 0, "Delta D for conservative revision must be 0"

obs_nonconservative = {"a": ("obs_X", "reveals_a"), "b": ("obs_X", "reveals_b")}
theta_nonconservative = theta_base
D_nonconservative   = compute_D(obs_nonconservative, theta_nonconservative)
assert D_nonconservative == 0, "Non-conservative revision must remove obstruction"
assert D_nonconservative - D_base == -1, "Delta D for non-conservative revision
    must be -1"

conservative_overturns      = 0
nonconservative_overturns   = 1
actual_conservative_overturns = 1 - D_conservative
actual_nonconservative_overturns = 1 - D_nonconservative
assert actual_conservative_overturns == conservative_overturns
assert actual_nonconservative_overturns == nonconservative_overturns

n_generators      = 10
m_hidden_states   = 100
revision_subsets  = 2 ** n_generators
hidden_pairs      = m_hidden_states * (m_hidden_states - 1) // 2

```

```

total_checks      = revision_subsets * hidden_pairs
assert revision_subsets == 1024
assert hidden_pairs == 4950
assert total_checks == 5_068_800

if __name__ == "__main__":
    print("All assertions passed.")
    print(f"Triangle: rank={triangle_rank}, H1={triangle_h1}, "
          f"global_sections={triangle_gs}, D={triangle_D}")
    print(f"Kernel-pair: H1={kernel_h1}, D={kernel_D}")
    print(f"H1-only recall={h1_recall} | Descent recall={descent_recall}")
    print(f"D_base={D_base}, D_conservative={D_conservative}, "
          f"D_nonconservative={D_nonconservative}")
    print(f"Conservative overturns={actual_conservative_overturns}, "
          f"Non-conservative overturns={actual_nonconservative_overturns}")
    print(f"Revision audit cost (n={n_generators}, m={m_hidden_states}): "
          f"{total_checks:,} pair-checks")

```